

1. Project Overview:

The purpose of this project is to analyze customer service performance using customer support ticket data. The analysis explores the distribution of ticket status, customer satisfaction ratings, ticket closure rates across different priority levels, and satisfaction across customer age groups. It also examines response-to-resolution times and identifies potential data-quality issues that could affect performance analysis.

2. Data Preparation

2.1 Missing Values:

The missing values in Resolution, Time to Resolution, and Customer Satisfaction Rating corresponded to non-closed tickets—specifically tickets with a status of **Open** or **Pending Customer Response**. All **Closed** tickets contained values in these fields. Therefore, these missing values were retained rather than automatically removed because they reflected the current status of the tickets.

2.2 Datetime Conversion:

First Response Time and Time to Resolution were converted from object to datetime format to enable time-based calculations. This allowed the time interval between the first response and recorded resolution to be calculated for each closed ticket.

2.3 Data Preparation: Age Group Creation

A new column, Age_group, was created from Customer Age by grouping customers into five age ranges. This made it easier to calculate and compare average customer satisfaction ratings across different age groups rather than analyzing each individual age separately.

2.4 Data Quality:

The analysis identified 1,365 negative response-to-resolution durations, representing approximately 49.3% of Closed tickets with recorded durations. These records were not removed from the dataset because they contained other valid information, including customer satisfaction ratings. Instead, the timestamp inconsistency was documented as a significant data-quality limitation, and only valid durations were used when analyzing response-to-resolution performance.

3. Key Findings and Visualizations.

3.1 Ticket Status Distribution

The customer support tickets were distributed across three categories: Pending Customer Response, Open, and Closed. Out of 8,469 tickets that were analyzed, 2,769 were Closed, representing approximately 32.7% of all the tickets. All Closed tickets had recorded

customer satisfaction ratings, while Open and Pending Customer Response tickets had no ratings recorded.

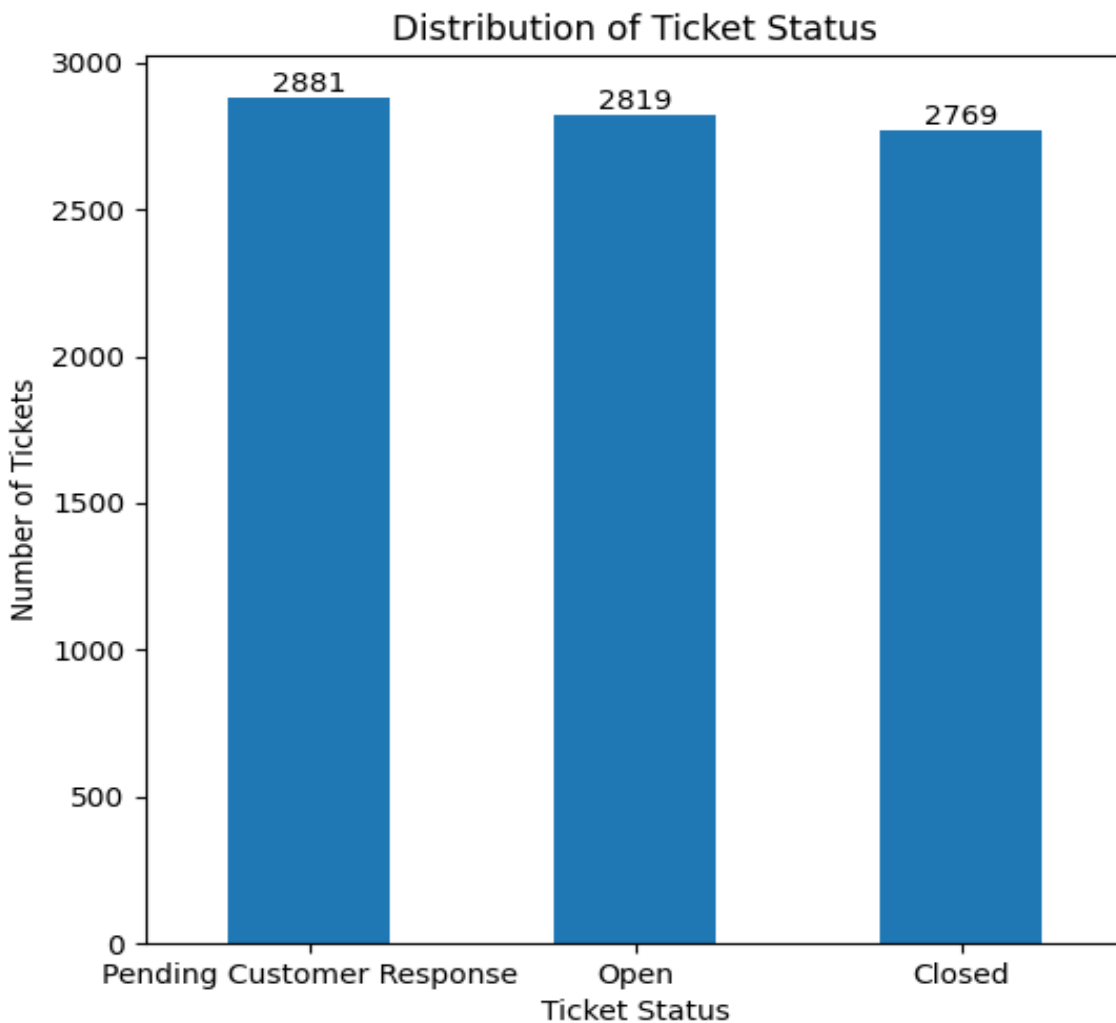


Figure 1: Distribution of ticket status

Business Recommendation: The customer service manager should investigate why approximately 67.3% of tickets remain Open or Pending Customer Response. This could involve reviewing ticket ages, identifying potential delays in resolution, and examining whether tickets are waiting for customer feedback or further action from support staff.

3.2 Customer Satisfaction Rating

The average customer satisfaction rating among Closed tickets was approximately 2.99 out of 5. Although the average customer satisfaction rating was 2.99 out of 5, the distribution showed that ratings were evenly spread across all five levels. Rating 3 was slightly more common, accounting for 20.95% of Closed tickets. This suggests that customer experiences varied rather than being concentrated at one satisfaction level.

Customer Satisfaction Rating	Number of Tickets	Percentage of Closed Tickets
1	553	19.97%
2	549	19.83%
3	580	20.95%
4	543	19.61%
5	544	19.65%

Figure 2: Customer satisfactory rating table

Business Recommendation:

The customer service manager should investigate the factors contributing to customer satisfaction ratings. This could involve reviewing customer feedback, examining recurring complaints, and identifying areas where customer service could be improved.

3.3 Ticket Closure Rate by Priority

Customer support tickets were classified into four priority levels: Critical, High, Medium, and Low. Critical-priority tickets had the highest closure rate at approximately 34.1%, followed closely by High-priority tickets at 33.81%. Medium- and Low-priority tickets had closure rates of 31.66% and 31.22%, respectively. Overall, closure rates were relatively similar across all four priority levels.

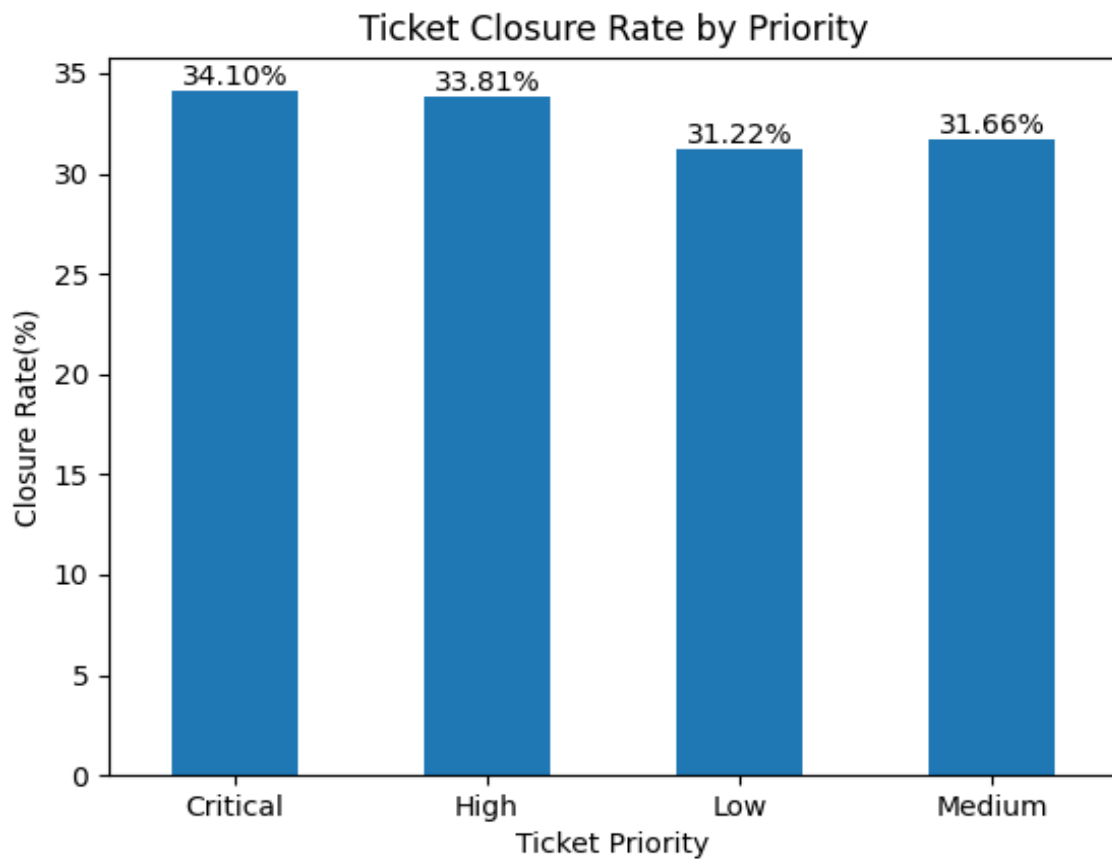


Figure 3: Ticket Closure Rate by Priority

Business Recommendation: The customer service manager should investigate why closure rates are relatively similar across the four priority levels. This could involve examining staffing levels, ticket assignment procedures, workload distribution, and whether Critical-priority tickets are being handled according to the company's established service-level targets.

3.4 Customer Satisfaction by Age Group

Customer satisfaction ratings were very similar across all five age groups. Customers aged 31–40 had the highest average satisfaction rating at approximately 3.03/5, while customers aged 51–60 had the lowest at approximately 2.94/5. The small difference of approximately 0.09 points suggests that customer satisfaction did not vary substantially across age groups in this dataset.

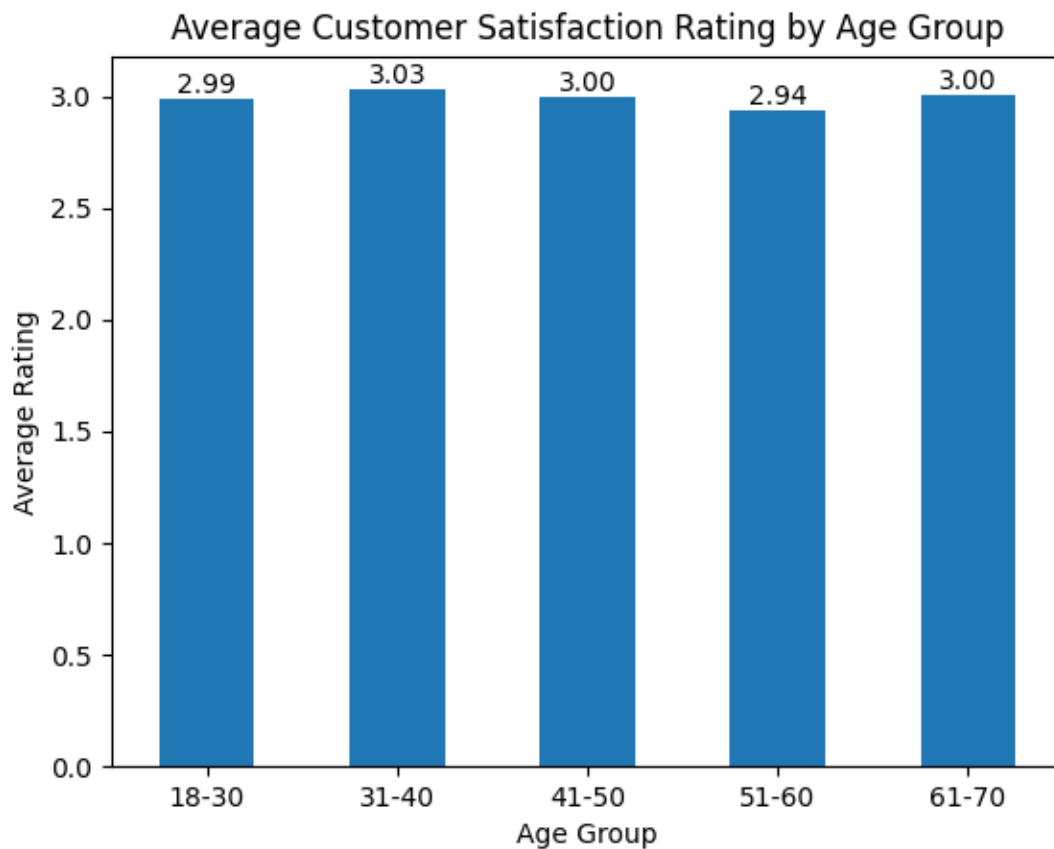


Figure 4: Average Customer Satisfaction Rating by Age Group

Business Recommendation: The customer service manager should investigate common service issues affecting customers across all age_groups. This is because average satisfaction ratings were relatively similar across all five age groups, suggesting that opportunities for service improvement may be shared across the customer base.

3.5 Response-to-Resolution Time and Data Quality

Out of 2,769 Closed tickets, 1,365 (49.3%) had negative response-to-resolution durations, meaning the recorded resolution timestamp occurred before the first-response timestamp. This indicates a significant data-quality issue that requires further investigation. Consequently, response-to-resolution performance metrics should be interpreted cautiously because almost half of the recorded durations were inconsistent.

Among tickets with valid positive durations, the average time between first response and recorded resolution was approximately 7 hours and 35 minutes. However, this average represents only the valid positive-duration subset, not all Closed tickets

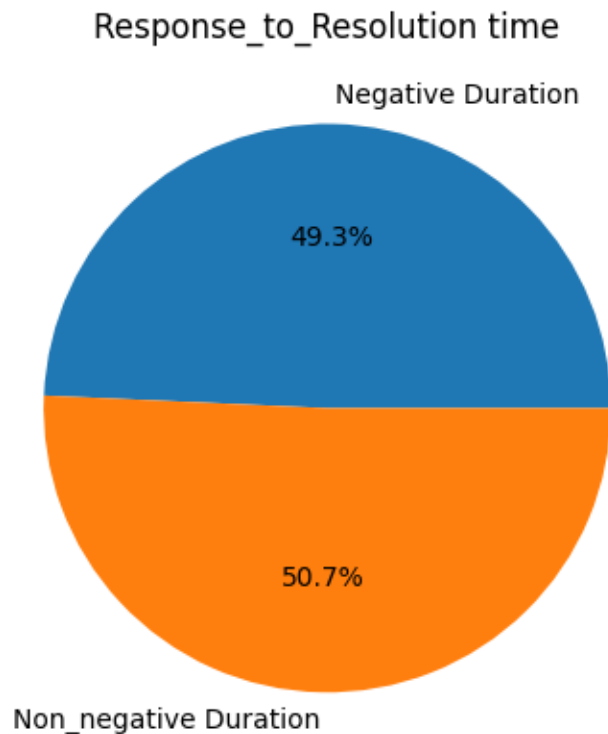


Figure 5: Response-to-Resolution Timestamp Validity

Business Recommendation: The customer service manager should investigate why some resolution timestamps were recorded before their corresponding first-response timestamps. The manager should also review why ticket creation timestamps were not included in the dataset. Ensuring that ticket creation, first response, and resolution timestamps are accurately recorded would improve data quality and enable more reliable measurement of customer service performance.

Limitation: The average response-to-resolution duration of approximately 7 hours and 35 minutes was calculated using only Closed tickets with positive durations. Because 49.3% of Closed tickets had inconsistent timestamp ordering, this average may not accurately represent the entire population of Closed tickets.

4. Project Conclusion

Out of 8,469 customer support tickets analyzed, only 32.7% were recorded as Closed, while the remaining 67.3% were Open or Pending Customer Response. This finding warrants further investigation into ticket ages, resolution targets, and possible operational delays to determine whether the proportion of unresolved tickets represents a customer service performance concern.

The analysis of 8,469 customer support tickets revealed that closure rates were relatively similar across the four ticket priority levels, while average customer satisfaction ratings showed little variation across age groups. No particular priority category or age group showed a substantial difference in its respective performance measure.

However, the analysis also identified a significant data-quality concern: 49.3% of Closed tickets had negative response-to-resolution durations. This limits the reliability of time-based performance analysis and highlights the need to improve timestamp recording and validation.